

**KEY**

The purple arrows show the direction of light rays. Black and blue arrows are nerve signals going to the optic nerve.

- ... → Light rays
- ... → Black and white
- ... → Color

Signal for black and white passes from retina to optic nerve

Ganglion cell

Signal for color passes from retina to optic nerve

Bipolar cell

3 The retina

The retina is made up of three layers. Light rays travel through the first two layers, ganglion and bipolar cells, and reach the third layer, which contains light-sensitive rod and cone cells. These convert light rays into nerve signals.

RETINA

Light ray travels to back of retina

Rods work in grayscale, responding to intensity of light; they enable us to see in dim conditions

Cones send nerve signals in response to green, red, or blue light; they need bright light to produce a signal

NERVE CELLS

LIGHT RECEPTOR CELLS

OPTIC NERVE

CHOROID

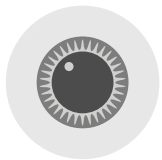
Wall of pigment cells forming back of retina

Optic nerve carries signals from light sensors to brain

OPTIC NERVE

4 Nerve signals to brain

The nerve signals trigger impulses in the ganglion and bipolar cells, which connect directly to the optic nerve. The nerve signals travel along the optic nerve to the brain.



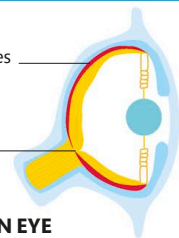
YOUR EYEBALLS REMAIN THE SAME SIZE THROUGHOUT YOUR LIFE

THE BLIND SPOT

To connect to the brain, the nerve fibers of the retina must pass through the back of the eye to form the optic nerve. This creates a "blind spot" that has no photoreceptors. We don't notice this because each eye provides data about a scene and the brain uses information from the other eye to complete the picture.

Rods and cones

Blind spot where nerve fibers leave eye



HUMAN EYE